Forecasting Asset Returns with Sentiment-Enhanced FIGARCH Models

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Abstract

This paper introduces a novel sentiment-driven framework for forecasting asset returns and modeling volatility by integrating machine learning and time-series econometric techniques. Leveraging a dataset of over 347,000 asset-specific news headlines and financial data spanning from March 2022 to January 2025, we first develop a Random Forest classifier to assess the likelihood that daily news headlines will significantly impact asset returns positively or negatively across four key time frames: Open-to-Close, Open-to-Open, Close-to-Close, and Close-to-Open. In parallel, we utilize the FinBERT model to extract average sentiment scores from headlines and construct sentiment-based exogenous variables. These features are incorporated into an extended FIGARCH$(1, 1)$ model to examine their predictive contributions to return volatility. Empirical results across 37 assets from stock, cryptocurrency, and commodity markets demonstrate that the inclusion of sentiment and impact probability features improves model performance (measured by AIC, BIC, and log-likelihood), particularly in time frames sensitive to market openings and overnight news. Our findings underscore the value of combining textual sentiment indicators produced by the FinBERT model, which is specifically trained on financial data, alongside directional time frame-based impact probabilities computed by the Random Forest model with advanced volatility modeling to enhance financial forecasting and support informed decision-making in dynamic markets.

Keywords:
Sentiment Analysis, Random Forest Model, FinBERT, FIGARCH, Returns Prediction

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Email addresses: (), ()

1. Introduction

Profitable financial market decisions largely rely on predicting future price movements of risky assets. Making such accurate predictions is difficult because they are influenced by many factors, and many of these factors and their effects on markets are still unknown, especially those that are qualitative or behavioral. The collective behavior of market participants, commonly called market sentiment, is a crucial factor in price movements and short-term deviations from assets' intrinsic value [?]. Studying, quantifying, and modeling market sentiment to improve the accuracy of asset-price models has become a significant challenge and area of research in this field.

Financial news platforms publish an extensive array of articles each day to report on global macroeconomic events and company-specific developments that significantly impact overall market trends and the price movements of individual companies. Market participants scan the released news and adjust their decisions based on the information they interpret. Consequently, market sentiment or the collective behavior of market participants can be shaped and influenced by the information or even the tone of financial news. Recently, the number of sources and availability of financial news data have grown significantly, making it essential for market participants to have an automated system for collecting, processing, and analyzing their impacts on financial assets to produce valuable insights that support decision-making processes [? ?].

Scholarly research increasingly explores how news and media sentiment influence financial markets, going beyond traditional fundamental analysis. Early work by Antweiler and Frank [?] analyzed Internet stock message boards, revealing that message activity predicts market volatility and has a statistically significant, albeit economically small, effect on stock returns, with disagreement among messages associated with increased trading volume. Complementing this, Baker and Wurgler [?] demonstrated that broad investor sentiment, derived from various market proxies, disproportionately affects stocks with subjective valuations and limited arbitrage opportunities, influencing the cross-section of returns. Further advancing the methodological landscape, Kelly and Ahmad [?] highlighted the importance of domain-specific dictionaries for extracting sentiment from financial news, showing that negative sentiment in financial news could predict next-day stock and crude oil returns, thereby enhancing trading strategies. More recently, with the advent of advanced natural language processing techniques, studies have leveraged BERT-based models for sentiment analysis. Case and Clements [?] found that financial news sentiment published pre-trading is predictive of daily S&P 500 price changes, while longer-term economic news sentiment exhibits a statistically significant negative relationship with monthly returns. Building on this, Costola et al. [?] utilized a financial market-adapted BERT model to examine COVID-19 news, concluding that a statistically significant positive relationship exists between news sentiment scores and S&P 500 market returns, with negative news serving as the primary driver of market expectations during crises and business news identified as a key sentiment driver. Specifically for crude oil markets, Sahut et al. [?] demonstrated that news-based sentiment, when integrated into machine learning forecasting models, significantly improved crude oil price prediction during periods of high volatility such as the COVID-19 pandemic, contrasting its influence in other crisis periods.

Numerous natural language processing (NLP) techniques have been proposed to extract sentiment metrics from financial news and examine their influence on asset price dynamics. Overall, sentiment analysis methods can be categorized into lexicon-based and machine learning-based approaches [?]. These methods quantify sentiment polarity and assign sentiment scores to textual content by leveraging historical data patterns.

Previous studies have compared lexicon-based approaches, which rely on predefined dictionaries of positive and negative words, with machine learning-based approaches trained on large financial text corpora. While lexicon-based methods provide interpretability and quick deployment, machine learning models, such as transformer-based models, tend to perform better at capturing complex sentiment patterns, especially within domain-specific language. Indeed, recent research further underscores the effectiveness of advanced computational methods in quantifying and leveraging market sentiment for financial forecasting [? ? ? ? ?]. Research shows that sentiment analysis used as a pre-processing step to extract informative features can significantly reduce feature dimensions and improve prediction accuracy for stock market trends, even outperforming deep learning models that rely on high-dimensional raw text features [? ?]. For example, SVM-based models combined with sentiment-related features, such as polarity and subjectivity extracted using NLP libraries, have demonstrated better performance and computational efficiency in predicting abnormal returns [?]. Likewise, the inclusion of sentiment scores as exogenous factors has been shown to improve the goodness-of-fit of regression models for forecasting stock opening prices, with polynomial autoregressions often providing a better fit than linear ones [?]. In the context of the volatile cryptocurrency market, Natural Language Processing (NLP) has been applied to convert news data into numerical sentiment data, and it has demonstrated that both positive and negative streams of information exert significant influence on the returns and volatility of Bitcoin futures, challenging efficient market hypotheses [?]. Furthermore, when comparing various deep learning and machine learning sentiment analysis models, transformer-based models such as BERT and their derivatives (e.g., RoBERTa, used by Pysentimento) have consistently shown better accuracy in sentiment tagging from social media content than traditional SVM and CNN-LSTM models in predicting cryptocurrency prices and directions [?]. Specialized domain sentiment models, such as SKEP on financial corpora, as well as novel review weighting methods, have also been shown to significantly improve short-term stock trend forecasts, clearly illustrating the effectiveness of feature engineering and specialized training for extracting fine-grained investor sentiment [?]. This collection of work collectively points to the growing complexity and increasingly important role that sentiment analysis plays in uncovering complex market configurations and enhancing forecasting capacity across a variety of financial assets.

The numerical outputs generated by these sentiment models have been incorporated into various financial forecasting frameworks as exogenous variables to improve their predictive accuracy and practical applicability [?]. Among machine learning-based models, FinBERT stands out as a specialized sentiment analysis framework trained specifically on financial texts, thereby achieving greater accuracy within the financial domain [?]. Beyond assessing sentiment metrics, a more critical question is how sentiment expressed during different time intervals affects price dynamics across various intraday time frames. The traditional view suggests that positive sentiment leads to positive returns, while negative sentiment results in negative returns. However, does this assumption hold consistently across all return horizons within a trading day for both positive and negative directions? Currently, there is limited research that explicitly explores this question [?].

Financial return time series exhibit special characteristics such as asymmetry, volatility clustering, and long-term memory, which require the development of models to capture these features. Traditional models such as GARCH [?] and its extensions have been widely used to capture volatility clustering in return series. However, these models often fall short in capturing long-memory effects, prompting the adoption of fractionally integrated

GARCH (FIGARCH) models [?]. FIGARCH is a well-known and robust model designed to represent financial time series by modeling volatility and providing accurate estimates. Recently, many studies have attempted to improve this model's performance by incorporating exogenous variables or considering regime-switching structures based on sentiment analysis.

Ampountolas [?] found Support Vector Regression (SVR) to be superior for short-run volatility forecasting of gold, cocoa, and S&P500 relative to ordinary GARCH models. Feng et al. [?] found that macroeconomic news sentiment significantly affects the states of Japanese stock volatility, as measured by the MRS-LMGARCH, with negative news increasing volatility and positive news decreasing it. Ho et al. [?] found that firm-specific news sentiment has a significant effect of reducing intraday asset volatility persistence, especially with negative news having the greatest impact, especially in high volatility states. Shi and Ho [?] established a Two-State Three-State FIGARCH (3S-FIGARCH) model incorporating Markov Regime-Switching to estimate economic volatility states, offering better long-memory estimates and structural change detection. Shi and Ho [?] developed an MRS-FIEGARCH model to show that macroeconomic and negative firm-specific news sentiment have strong effects on volatility states, increasing the likelihood of higher volatility. Xu et al. [?] further found that incorporating investor sentiment and regime switching significantly improves the forecasting performance of realized range-based volatility (RRV) for the S&P 500 index. Xu et al. [?] also quantified Chinese official media sentiment (with BERT) as being important and negatively forecasting short-to-long horizon Chinese crude oil futures (SC) but not WTI volatility, indicating Chinese market-specific information.

Despite advances in volatility models and sentiment analytics, to the best of our knowledge, no research has focused on improving the FIGARCH model for return series estimation while considering the potentially different effects of financial news headlines on price dynamics across different directions and multiple time frames. This paper aims to bridge this gap by developing a sentiment-driven FIGARCH framework informed by a Random Forest-based news impact probabilities.

In this paper, we have developed a machine learning-based framework using the Random Forest model [?] to assess the likelihood that news headlines affect asset daily returns across different time frames, including Open-to-Close, Open-to-Open, Close-to-Open, and Close-to-Close, in both positive and negative directions. In this framework, news headlines collected within a day are classified as impactful or non-impactful based on their effect on return deviations. If published news during a day causes significant fluctuations in returns, we classify them as impactful, and vice versa. We have incorporated the bag-of-words feature representation into the Random Forest model to compute the probability that news headlines are impactful on returns across various time frames. Additionally, we used impact probabilities alongside sentiment variables derived from FinBERT scores, and we included news frequency as an exogenous variable in the FIGARCH time-series prediction model to examine their contributions to predictive accuracy and goodness-of-fit. We also compared the performance of our method through empirical studies on financial assets from different markets. A comprehensive dataset was compiled, including 347,500 asset-specific news headlines and financial data from March 1, 2022, to January 31, 2025, covering 36 assets from the stock, cryptocurrency, and commodity markets. The results indicate that, using the Random Forest model, we can accurately classify news headlines as impactful or non-impactful in the subsequent period. However, the accuracy varies across different time frames and directions; on average, the model for classifying positive impacts over the Close-to-Open time frame achieves the highest accuracy. Furthermore, the FIGARCH model with exogenous variables such as sentiment metrics and impact probabilities outperforms traditional FIGARCH models in terms of data fit. Therefore, our proposed sentiment analysis framework could serve as a broadly useful tool for modeling financial volatility and enhancing prediction accuracy.

This paper contributes to the literature in three ways. First, it introduces a novel sentiment-enhanced FIGARCH framework that integrates machine-learning–based news-impact probabilities with FinBERT-derived sentiment scores. Second, it explicitly examines how news sentiment affects returns across four distinct intraday horizons (Open-to-Close, Open-to-Open, Close-to-Close, and Close-to-Open), an aspect that has received little attention in prior FIGARCH research. Third, by evaluating 37 assets from stock, cryptocurrency, and commodity markets, the study provides comprehensive empirical evidence of the value of sentiment-driven exogenous variables in improving volatility modeling and forecasting.

The remainder of this paper is organized as follows. Section 2 describes the collected news and financial data, as well as the methodology and models implemented in this study. Section 3 discusses the empirical results. Finally, Section 4 presents the conclusion and directions for future research.

2. Methodology

2.1. Data

Our proposed approach is based on comprehensive data obtained from AlphaVantage [?] covering the period from March 1, 2022, to January 31, 2025. This dataset consists of asset-specific news headlines and financial data for a broad range of assets across the stock, commodity, and cryptocurrency markets, including the 23 largest companies in the Dow Jones Industrial Index (DJI), the 12 largest companies in the NASDAQ index, Brent Oil, and the two largest cryptocurrencies, Bitcoin and Ethereum. AlphaVantage aggregates news headlines from premier global news outlets related to specific firms or topics. The collected news headlines pertain to the stock and cryptocurrency markets, while those related to commodities cover macroeconomic issues, monetary policy, and energy transportation topics. AlphaVantage provides both the headlines and article summaries, along with their precise publication times from the outlet that reports the news before any other sources. Therefore, the collected headlines are considered novel, and it is assumed that the effect of the news on prices has not yet been reflected at the time of publication. Table 1 summarizes the number of headlines collected for each market.

Table 1: Overview of individual assets and the corresponding number of news headlines collected from AlphaVantage for each market.

Market	Assets	# of News Headlines
Stock	AAPL, AMGN, AMZN, AXP, AVGO, BA, CAT, COST, CRM, CSCO, CVX, DIS, GOOG, GS, HD, HON, IBM, JNJ, KO, META, MCD, MMM, MSFT, NFLX, NKE, NVDA, PG, SHW, TRV, TSLA, UNH, V, VZ, WMT	307,200
Cryptocurrency	Bitcoin, Ethereum	49,972
Commodity	Brent Oil	71,747

2.2. Sentiment Variables from FinBERT

In this research, we utilized the FinBERT model API, integrated by the Hugging Face community [?], to analyze the sentiment polarity of each news headline and to evaluate the corresponding sentiment scores and the frequencies of headlines in each sentiment class. Before extracting sentiment, noise in the collected news headlines was removed. Specifically, we removed common prefixes and suffixes in headlines that do not convey analytical or sentiment-bearing information. For instance, stylistic or source markers such as "Breaking:", "Update:", and "Report:", and publisher identifiers (e.g., "Reuters –" and "CNBC:"), as well as redundant stock ticker tags placed in parentheses (e.g., "(AAPL)" and "(TSLA)"). These terms, identified via manual inspection of numerous headlines, were discarded as they pertain solely to formatting or metadata rather than to the semantic content necessary for sentiment analysis. Furthermore, some texts mistakenly classified as headlines due to data-collection errors were identified and removed.

Using the positive, negative, and neutral sentiment scores, along with the frequency of polarized headlines provided by the FinBERT model, we constructed several variables for each trading day. These variables are used as exogenous variables in the models in subsequent steps to evaluate the effect of news sentiment on price predictions. They are defined as follows:

$$AS_t^+ = \frac{\sum_{i=1}^{N_t^+} S_{t,i}^+}{N_t^+},\ AS_t^- = \frac{\sum_{i=1}^{N_t^-} S_{t,i}^-}{N_t^-},\ AS_t^O = \frac{\sum_{i=1}^{N_t^O} S_{t,i}^O}{N_t^O}, \tag{1}$$

where $S_{t,i}^+$, $S_{t,i}^-$, and $S_{t,i}^O$ are, respectively, positive, negative, and neutral sentiment scores of the ith news headline at time t. N_t^+, N_t^-, and N_t^O are, respectively, the number of positive, negative, and neutral news headlines at time t.

2.3. Bag-of-Words Model

As discussed in the introduction, asset-specific news headlines may be impactful on financial returns in some time frames but not impactful in others. For instance, they may push up the Open-to-Close return of a particular asset but pull down its Close-to-Close return. Generally, collected news headlines related to any financial asset on a given day can be classified into three main categories: positively impactful, negatively impactful, and non-impactful on financial returns. In this part, we fit the Random Forest model using the generated bag-of-words from news headlines as features to

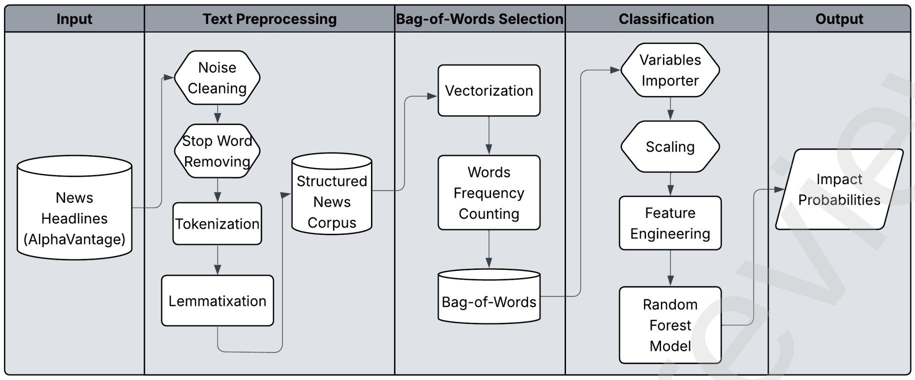

Figure 1: A system diagram of the classification model developed to compute time series of news headlines' impact probabilities.

calculate the probabilities that collected news headlines are positively or negatively impactful on different financial returns. The overall architecture of this model is shown in Figure 1.

We define four types of returns according to their time frames as follows:

$$r_{OC,t} = \ln\left(\frac{\mathrm{Close}_t}{\mathrm{Open}_t}\right), \qquad r_{OO,t} = \ln\left(\frac{\mathrm{Open}_t}{\mathrm{Open}_{t-1}}\right),$$
$$r_{CO,t} = \ln\left(\frac{\mathrm{Open}_t}{\mathrm{Close}_{t-1}}\right), \qquad r_{CC,t} = \ln\left(\frac{\mathrm{Close}_t}{\mathrm{Close}_{t-1}}\right),$$

where $r_{OC,t}$, $r_{OO,t}$, $r_{CO,t}$, and $r_{CC,t}$ represent, respectively, the Open-to-Close, Open-to-Open, Close-to-Open, and Close-to-Close returns at time t (measured in trading days). Furthermore, Open_t and Close_t denote the opening and closing prices at time t, respectively.

The binary response variables for the classification problems are defined as follows:

$$I^{+}_{w,t} = \begin{cases} 0 & \text{if } r_{w,t} \leq Q_{0.75}(r_w), \\ 1 & \text{if } r_{w,t} > Q_{0.75}(r_w), \end{cases} \tag{2}$$

$$I^{-}_{w,t} = \begin{cases} 0 & \text{if } r_{w,t} \geq Q_{0.25}(r_w), \\ 1 & \text{if } r_{w,t} < Q_{0.25}(r_w), \end{cases} \tag{3}$$

where w denotes the return time window, such that $w \in \{OC, OO, CO, CC\}$, and $Q_{0.75}(r_w)$ and $Q_{0.25}(r_w)$ represent the 75th and 25th percentiles of the entire sample of returns, r_w, with the time frame w, respectively. As follows from (2) and (3), the returns above the 75th percentile are considered significantly positive, while the returns below the 25th percentile are considered significantly negative.

To generate the bag-of-words dataset for the Random Forest model, we first removed noise and stop words from the news headlines. After tokenizing the headlines, parts of speech were detected using the `spacy 3.6` library in Python, and lemmatization was performed to extract word roots. In the next step, the tokens were converted to vectors using the Term Frequency–Inverse Document Frequency (TF-IDF) vectorizer from the *scikit-learn* library in Python, which measures not only how frequently a word appears in a document, but also how rare the word is across all documents. This technique helps reduce the impact of common words while emphasizing distinctive keywords.

To fit the Random Forest model, we split each asset's dataset into training and testing sets, with 80% of the samples used for training and 20% for testing. The model's hyperparameters were tuned using cross-validation. We fit eight distinct Random Forest models for each asset, each corresponding to a different response variable. The outputs for each asset, which can be considered predictive insights for the next steps, are defined as follows:

$$IP^{+}_{w,t} = \mathbb{P}(I^{+}_{w,t} = 1) = \mathbb{P}(r_{w,t} > Q_{0.75}(r_w)), \tag{4}$$
$$IP^{-}_{w,t} = \mathbb{P}(I^{-}_{w,t} = 1) = \mathbb{P}(r_{w,t} < Q_{0.25}(r_w)). \tag{5}$$

In other words, for each asset and for each time frame w, we build and train two Random Forest classification models that estimate the probabilities $IP^{-}_{w,t}$ and $IP^{+}_{w,t}$ that the news headlines posted over the news collection window (as shown in Figure 2) have, respectively, negative and positive impacts on the asset return $r_{w,t}$. These estimates are then used as exogenous variables in the return forecasting model, as explained in the next section.

To make our approach practically feasible, the news data used for the response variables correspond to a specific time interval preceding the return prediction window. In particular, we use the news data from the previous 24 hours before the relevant return window. This rule is illustrated in Figure 2.

Performance metrics of the Random Forest classification models across different return horizons and positive/negative impact classifications are summarized in Table 2. Each metric is averaged over all investigated financial

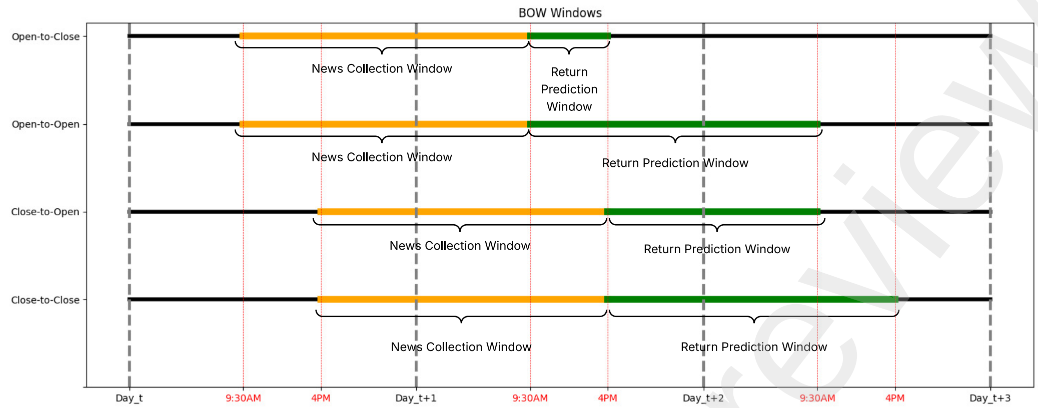

Figure 2: Data collection and return prediction windows for each of the Open-to-Close ($r_{OC,t}$), Open-to-Open ($r_{OO,t}$), Close-to-Open ($r_{CO,t}$), and Close-to-Close ($r_{CC,t}$) returns.

assets. In the test sets, observations with predicted probabilities of 0.25 or higher were classified as belonging to the impactful class, while those below this threshold were assigned to the non-impactful class. The threshold value was selected to maximize the average performance across both classes in an imbalanced dataset where non-impactful news headlines dominate. The performance metrics include overall accuracy, F1-scores for both impactful and non-impactful news, and ROC-AUC values.

Table 2: Average performance metrics of Random Forest classification models for predicting positively and negatively impactful news across different return time frames.

Average Metric	Response Variable							
	$I^+_{OC,t}$	$I^-_{OC,t}$	$I^+_{OO,t}$	$I^-_{OO,t}$	$I^+_{CO,t}$	$I^-_{CO,t}$	$I^+_{CC,t}$	$I^-_{CC,t}$
`Accuracy`	0.905	0.895	0.907	0.898	0.901	0.888	0.915	0.903
`F1` Impactful	0.813	0.796	0.815	0.796	0.809	0.783	0.835	0.807
`F1` Non-Impactful	0.936	0.929	0.938	0.932	0.933	0.924	0.943	0.935
`ROC-AUC`	0.937	0.935	0.930	0.925	0.940	0.933	0.946	0.940

The evaluation metrics used here are as follows. `Accuracy` is the proportion of instances correctly classified and is computed as:

$$\texttt{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{6}$$

where TP, TN, FP, and FN stand for true positives, true negatives, false positives, and false negatives, respectively.

`F1-score` is the harmonic mean of recall and precision, and for the positive (impactful) class is computed as:

$$\texttt{F1-score} = 2 \cdot \frac{\texttt{Precision} \cdot \texttt{Recall}}{\texttt{Precision} + \texttt{Recall}}, \tag{7}$$

where $\texttt{Precision} = \dfrac{TP}{TP + FP}$ and $\texttt{Recall} = \dfrac{TP}{TP + FN}$.

`ROC-AUC` (Receiver Operating Characteristic – Area Under the Curve) is the area under the ROC curve, obtained by plotting the true positive rate $TPR = \dfrac{TP}{TP + FN}$ vs. the false positive rate $FPR = \dfrac{FP}{FP + TN}$ at various threshold settings. Greater `ROC-AUC` indicates superior discriminative performance.

The probability threshold of 0.25 balances classification performance across classes in the presence of imbalance. While threshold tuning can improve precision–recall trade-offs, the primary objective of our classification stage is not hard-label prediction but rather the generation of impact probabilities, which are subsequently incorporated as exogenous variables in the FIGARCH models. Within this probabilistic framework, the high ROC-AUC values reported in Table 2 confirm that the classifier possesses strong discriminative ability, ensuring that the probability estimates are reliable inputs for the subsequent volatility modeling.

2.4. FIGARCH Return Prediction Model

In this section, we have adjusted the return prediction FIGARCH$(1, 1)$ model by sentiment-driven exogenous sets of variables derived in Sections 2.2 and 2.3. The FIGARCH model, introduced by Baillie et al. [**?**], extends the GARCH family of models. Beyond capturing volatility clustering and providing reliable in-sample estimates [**? ?**], the FIGARCH model is specifically designed to account for the long-memory characteristics observed in financial volatility. The auto-regressive mean equation is modified to our approach by including sentiment-driven exogenous variables. The augmented

FIGARCH(p,q) model is described as follows:

$$\begin{aligned} r_t &= \mu + \sum_{i=1}^{\ell} \alpha_i r_{t-i} + \boldsymbol{\gamma} \mathbf{X}_t^\top + \varepsilon_t, \\ \varepsilon_t &= \sigma_t \cdot z_t, \quad z_t \sim t(0,1,\nu), \\ \sigma_t^2 &= \omega + \sum_{i=1}^{q} \phi_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2 + \left(\sum_{i=1}^{q} \phi_i \varepsilon_{t-i}^2 \right) (1-L)^d, \end{aligned} \tag{8}$$

where r_t is log-return at time t, μ is expected log-return for the asset, α_i is the coefficient of ith auto-regressive (AR) term, $\mathbf{X}_t^\top$ is the vector of exogenous variables at time t, $\boldsymbol{\gamma}$ is the corresponding coefficients vector, σ_t^2 is the conditional volatility at time t, ε_t is the shock term at time t, $z_t \sim t(0,1,\nu_{S_t})$ is a standardized Student's t-distributed innovation term with mean 0, variance 1, and ν degrees of freedom, ω is the constant term in the volatility equation, α is the impact of past squared shocks (ARCH term), β is the persistence of past volatility (GARCH term), L is the lag operator, and d is the fractional differencing parameter.

In what follows, we set $p = 1$, $q = 1$, and $\ell = 1$ in (8). This choice is based on both theoretical and empirical reasons. From a modeling standpoint, the FIGARCH$(1,1)$ framework is seen as the most efficient and empirically reliable option, capturing the key long-memory features of financial volatility without adding unnecessary complexity.

Return prediction time-series models typically assume that the error terms follow a normal distribution with stationary deviations, enabling the use of the ordinary least squares (OLS) method to estimate parameters [?]. In this study, we have considered a fat-tailed distribution, specifically Student's t-distribution, to better fit the data samples.

The parameters of the FIGARCH model were estimated using maximum likelihood estimation (MLE). We employed the `arch 7.2.0` Python library, which optimizes the joint log-likelihood function of the conditional mean equation (including AR terms and sentiment-driven exogenous variables) and the time-varying variance equation. The optimization was performed using the Sequential Least Squares Programming (SLSQP) algorithm implemented in the `scipy.optimize v1.15.3` module. This algorithm enforces standard parameter constraints, including bound restrictions on the fractional differencing parameter ($d \in [0,1]$), non-negativity constraints on the volatility equation parameters ($\omega \geq 0$, $\phi_i \geq 0$, $\beta_j \geq 0$), and stationarity constraints for the mean process. In our implementation, estimation reliability was further ensured by relying on the convergence diagnostics provided by the `arch 7.2.0` library. Robust covariance estimation (`cov_type="robust"`) is computed only when the optimizer converges successfully, which guarantees that the reported results are based on valid solutions. The `arch_model.fit()` routine also provides detailed convergence information, including the optimizer's exit status, the number of iterations, and the number of function evaluations. In addition, our code explicitly excluded any cases where convergence was not achieved through error handling, ensuring that all results presented in this study are based exclusively on successfully converged models without parameter boundary violations.

3. Empirical Results

In this section, we present the empirical evaluation of the proposed sentiment-driven FIGARCH$(1, 1)$ model for return prediction. The objective is to assess the predictive power of incorporating news-based sentiment metrics and impact probabilities into financial time-series modeling. Using a dataset that includes 347,500 asset-specific news headlines and corresponding financial data from March 1, 2022, to January 31, 2025, we conduct a series of experiments across 36 assets from the stock, cryptocurrency, and commodity markets using the entire dataset.

For each asset, we begin by defining the sets of exogenous variables produced in Sections 2.2 and 2.3. The time series from Section 2.2 are the proposed average sentiment scores for each group of positive, negative, and neutral daily published news headlines. The variables from Section 2.3 are probability time series representing the likelihood that positive or negative news headlines influence financial returns across various time frames. After verifying the stationarity of all exogenous time series using the Dickey–Fuller unit root test, we fit augmented FIGARCH models that incorporate these exogenous variables, alongside a traditional FIGARCH model used as a benchmark. We then evaluate their performance by comparing goodness-of-fit metrics and predictive accuracy across the full dataset for different assets and return horizons. In addition, we examine the proportion of statistically significant coefficients across model specifications. The augmented FIGARCH$(1, 1)$ model and sets of exogenous variables are defined as follows:

$$
\begin{aligned}
r_{w,t} &= \mu_w + \alpha_w \; r_{w,t-1} + \boldsymbol{\gamma}\mathbf{X}_t^{s\top} + \varepsilon_t \\
\varepsilon_t &= \sigma_t \cdot z_t, \quad z_t \sim t(0,1,\nu), \\
\sigma_t^2 &= \omega + \phi_1\varepsilon_{t-1}^2 + \beta_1\sigma_{t-1}^2 + \left(\phi_1\varepsilon_{t-1}^2\right)(1-L)^d,
\end{aligned} \tag{9}
$$

where

$$
\begin{aligned}
&\mathbf{X}^1 = \{IP_{OC}^+\ , IP_{OC}^-\}, \\
&\mathbf{X}^2 = \{IP_{OO}^+\ , IP_{OO}^-\}, \\
&\mathbf{X}^3 = \{IP_{CC}^+\ , IP_{CC}^-\}, \\
&\mathbf{X}^4 = \{IP_{CO}^+\ , IP_{CO}^-\}, \\
&\mathbf{X}^5 = \{IP_{OC}^+\ , IP_{OC}^-\ , IP_{OO}^+\ , IP_{OO}^-\ , IP_{CC}^+\ , IP_{CC}^-\ , IP_{CO}^+\ , IP_{CO}^-\}, \\
&\mathbf{X}^6 = \{AS_t^+,\ AS_t^-,\ AS_t^O\}, \\
&\mathbf{X}^7 = \{IP_{OC}^+\ , IP_{OC}^-,\ IP_{OO}^+\ , IP_{OO}^-,\ AS_t^+,\ AS_t^-,\ AS_t^O\}, \\
&\mathbf{X}^8 = \{IP_{OC}^+\ , IP_{OC}^-\ , IP_{OO}^+\ , IP_{OO}^-\ , IP_{CC}^+\ , IP_{CC}^-\ , IP_{CO}^+\ , IP_{CO}^-, \\
&\qquad\quad AS_t^+,\ AS_t^-,\ AS_t^O\},
\end{aligned}
$$

and

$$
s \in \begin{cases} \{1,2,6,7\} & \text{if } w \in \{OC, OO\}, \\ \{1,2,\ldots,8\} & \text{if } w \in \{CC, CO\}. \end{cases} \tag{10}
$$

The parameter s defined in (10) determines the exogenous set included in the model based on the prediction time frame. The assumption is that for models predicting returns in Open-to-Close and Open-to-Open time frames, we do not use impact probability variables for Close-to-Close and Close-to-Open time frames. The reason is that the data collection window for computing the variables in the Close-to-Close and Close-to-Open time frames overlaps with the prediction window for returns in the Open-to-Close and Open-to-Open time frames. In other words, for predicting returns in a specific period of time, we do not use any information from that period itself, and all information used for forecasting comes from past periods.

The augmented model in (9) is fitted for all 37 assets in each of the four time frames, as well as for every set of exogenous variables (8 exogenous sets) in addition to the benchmark FIGARCH model without exogenous variables (the base model). Considering that we only have Open-to-Open data for Brent Oil in the commodity market, in total, we have estimated parameters for a total of 1011 variants, including the augmented model in (9) and the benchmark model obtained by setting $\boldsymbol{\gamma}$ in (9) equal to zero.

Table 3 summarizes the number of times each coefficient was statistically significant across all model estimations, considering various sets of exogenous variables and time frames. Among the core FIGARCH parameters, the degrees of freedom parameter (ν) and the long-memory component (d) exhibit the highest levels of significance, with ν significant at the 1% level in 689 out of 1000 estimations. The GARCH term (β) is also frequently significant, while the ARCH term (ϕ) shows relatively limited significance, indicating that volatility persistence plays a more critical role than short-term shocks. Regarding the exogenous variables, the impact probability features ($IP^{+}_{OO}, IP^{-}_{OC}, IP^{+}_{OC}$, etc.) demonstrate strong statistical relevance, particularly in the Open-to-Open and Close-to-Open return windows. Sentiment score aggregates (AS^{+} and AS^{-}) are also frequently significant, whereas neutral sentiment coefficients (AS^{O}) were rarely significant across model specifications (only 19 out of 353 at the 1% level). Nevertheless, we included neutral sentiment to ensure transparency and robustness of the analysis. Since the FinBERT model produces positive, negative, and neutral sentiment scores, evaluating all categories avoids potential bias from selective exclusion of variables ex ante. Empirically, our findings confirm that neutral sentiment provides little explanatory power compared to positive and negative sentiment, which is consistent with the theoretical expectation that market behavior is primarily driven by polarized sentiments. These results highlight the importance of incorporating sentiment into volatility modeling frameworks. Furthermore, sentiment with positive and negative impacts may exhibit different predictive power, and their contributions to volatility models vary depending on the time frame.

Table 3: Number of statistically significant coefficients at different confidence levels across models estimated for all assets, using various sets of exogenous variables in the Open-to-Open, Open-to-Close, Close-to-Close, and Close-to-Open time frames. For each coefficient, the table shows the total number of times it was included and how often it was statistically significant at the 10%, 5%, and 1% confidence levels, respectively.

Coefficient	**Number of models**	**Confidence Level**		
		10%	**5%**	**1%**
μ	1000	582 (58.2%)	534 (53.4%)	479 (47.9%)
ω	1000	430 (43.0%)	383 (38.3%)	330 (33.0%)
ϕ	1000	105 (10.5%)	86 (8.6%)	69 (6.9%)
d	1000	619 (61.9%)	582 (58.2%)	514 (51.4%)
β	1000	570 (57.0%)	540 (54.0%)	490 (49.0%)
ν	1000	784 (78.4%)	761 (76.1%)	689 (68.9%)
α_{OC}	144	50 (34.7%)	46 (31.9%)	43 (29.9%)
α_{OO}	140	36 (25.7%)	35 (25.0%)	31 (22.1%)
α_{CC}	287	107 (37.3%)	101 (35.2%)	88 (30.7%)
α_{CO}	285	81 (28.4%)	77 (27.0%)	64 (22.5%)
IP^{+}_{OC}	424	212 (50.0%)	184 (43.4%)	150 (35.4%)
IP^{-}_{OC}	424	213 (50.2%)	197 (46.5%)	156 (36.8%)
IP^{+}_{OO}	423	246 (58.2%)	228 (53.9%)	193 (45.6%)
IP^{-}_{OC}	423	256 (60.5%)	239 (56.5%)	207 (48.9%)
IP^{+}_{CC}	216	121 (56.0%)	115 (53.2%)	100 (46.3%)
IP^{-}_{CC}	216	120 (55.6%)	108 (50.0%)	86 (39.8%)
IP^{+}_{CO}	216	123 (56.9%)	111 (51.4%)	93 (43.1%)
IP^{-}_{CO}	216	110 (50.9%)	101 (46.8%)	91 (42.1%)
AS^{+}	353	190 (53.8%)	177 (50.1%)	156 (44.2%)
AS^{O}	353	24 (6.8%)	23 (6.5%)	19 (5.4%)
AS^{-}	353	212 (60.1%)	188 (53.3%)	169 (47.9%)

To evaluate the predictive performance of the proposed FIGARCH models, we estimate separate models that incorporate various sets of sentiment-driven exogenous variables, $\mathbf{X}^1, \ldots, \mathbf{X}^8$. These models are compared against a benchmark FIGARCH model without any exogenous inputs (denoted as **Base**) to determine the added value of including external information. The evaluation relies on three standard performance metrics: the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and the Log-Likelihood Function (LLF). Both AIC and BIC are likelihood-based criteria that balance model fit and complexity by penalizing the inclusion of extra variables, with lower values indicating better performance. The LLF, on the other hand, measures the likelihood of observing the data given the model parameters, with higher values signifying a better fit. Collectively, these criteria provide a comprehensive assessment of model quality, capturing both in-sample fit and parsimony.

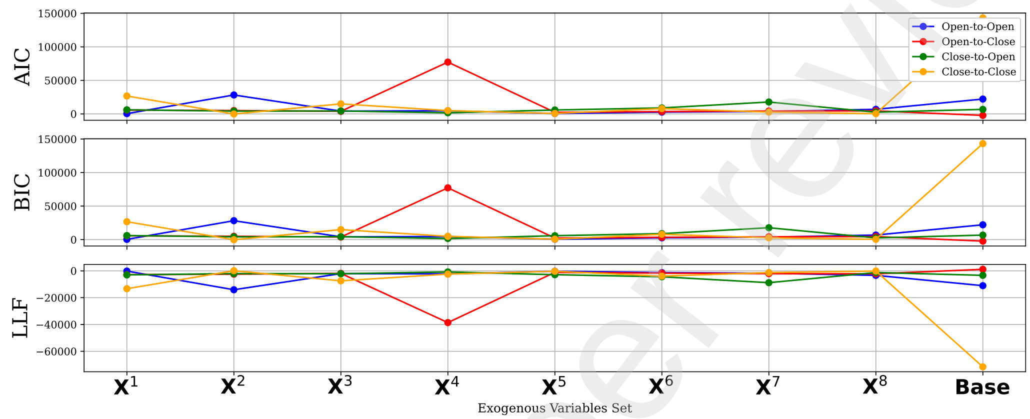

Figure 3: Model performance across different exogenous variable sets for stocks. The metrics include average AIC, average BIC, and average LLF. Lower AIC/BIC and higher LLF values indicate better model fit.

Figure 3 illustrates the performance of the FIGARCH model across various sets of exogenous variables corresponding to stock market assets. For the Close-to-Close time frame, models incorporating variables from all exogenous sets show significant improvements over the benchmark **Base** model for assets in the stock market. However, nearly all models with sentiment-driven exogenous variables (except the one that incorporates $\mathbf{X}^4$ in the Open-to-Close time frame) outperform the benchmark model in the stock market.

In the Open-to-Close setting shown in Figure 4, which illustrates the performance of models in the cryptocurrency market, all variable sets except the set $\mathbf{X}^2$ produce lower AIC and higher LLF than the benchmark model, indicating strong predictive value from sentiment-driven features. The same applies in the Close-to-Open setting, where $\mathbf{X}^1$ is excluded. However, cryptocurrency models show relatively flat performance across all variable sets, suggesting that sentiment signals may be less informative in this market.

Figure 5 illustrates the performance of FIGARCH models for Brent Oil across different exogenous variable sets ($\mathbf{X}^2$, $\mathbf{X}^6$, and **Base**) under the Open-to-Open time frame. Both AIC and BIC show notable improvements when

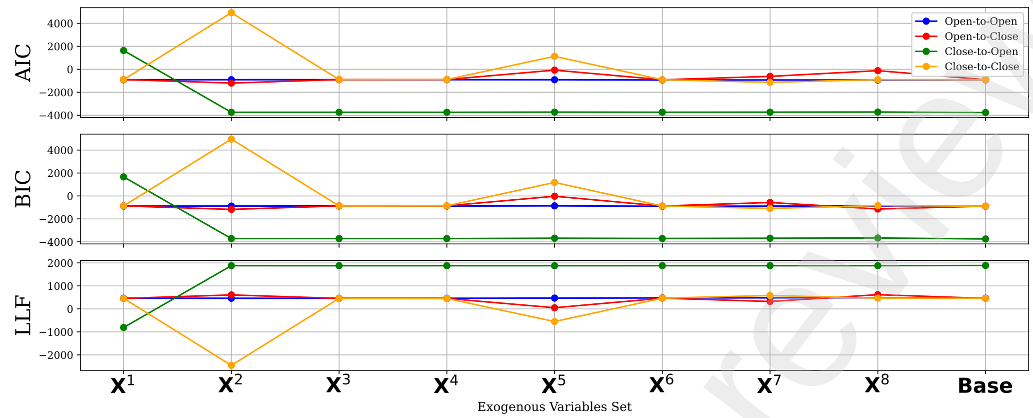

Figure 4: Model performance across different exogenous variable sets for cryptocurrency assets. Average AIC, BIC, and LLF are used as model selection criteria.

sentiment-based exogenous variables are included, with $\mathbf{X}^2$ achieving the largest reduction, indicating its stronger contribution to model fit. Similarly, LLF values are consistently higher for augmented models compared to the benchmark (**Base**), further confirming the enhanced explanatory power of sentiment information. Overall, the results demonstrate that incorporating sentiment-derived exogenous variables substantially improves the goodness-of-fit of FIGARCH models in the Brent Oil market, even though the analysis is restricted to the Open-to-Open horizon due to data availability.

For completeness and ease of reference, the numerical results corresponding to the figures presented in this section are reported in Tables A.4–A.7 in Appendix Appendix A. These tables provide the full set of model performance metrics across assets, return horizons, and exogenous variable sets, thereby facilitating detailed inspection and ensuring transparency and reproducibility of the findings.

4. Conclusion

This study proposed a sentiment-aware framework for predicting financial returns by integrating machine learning classification and volatility modeling techniques. Specifically, we employed a Random Forest classifier to assess the likelihood that news headlines affect asset returns across multiple time frames and used FinBERT sentiment scores to construct interpretable sentiment variables. These features were incorporated as exogenous variables

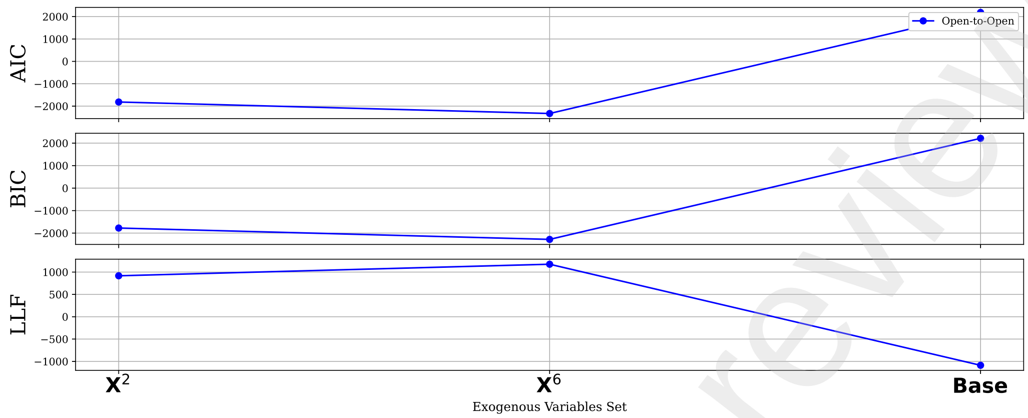

Figure 5: Model performance across different exogenous variable sets for commodity market (Brent Oil). Brent Oil is only investigated in Open-to-Open time frame due to data availability. AIC, BIC, and LLF are used as model selection criteria.

into an extended FIGARCH(1,1) model to assess their influence on volatility and return dynamics. Empirical evaluations on a large-scale dataset covering 37 financial assets from stock, cryptocurrency, and commodity markets demonstrated that incorporating sentiment-driven variables improves model performance in terms of fit and explanatory power, especially in Open-to-Open and Close-to-Open time frames. Our results highlight the predictive value of combining textual sentiment indicators with time-series modeling for better capturing the long-memory behavior of financial volatility.

Beyond statistical improvements, the results have clear financial implications. The significance of negative sentiment and negative impact probabilities, particularly in the Open-to-Open and Close-to-Open horizons, supports theories of asymmetric volatility and loss aversion, where adverse news triggers disproportionately strong market reactions. The stronger predictive power of overnight sentiment variables is consistent with market microstructure research, which shows that information released after trading hours is abruptly incorporated at the market's opening, resulting in volatility shocks. At the same time, the negligible role of neutral sentiment confirms that markets respond primarily to salient signals rather than ambiguous news, consistent with attention-based trading theories. Finally, differences in the predictive power of sentiment variables across asset classes, strong for equities, weaker for cryptocurrencies, and macro-driven for commodities, illustrate how sentiment interacts with the informational structures specific to each market. These insights highlight the economic significance of our findings, demonstrating that sentiment-enhanced FIGARCH models not only improve statistical fit but also reveal behavioral mechanisms underlying asset volatility.

This research provides a promising tool for market participants aiming to integrate qualitative information into quantitative forecasts. The primary contribution of this study is to demonstrate how sentiment-derived variables can be systematically integrated into FIGARCH models to improve volatility forecasting across multiple return horizons. Unlike much of the recent literature that emphasizes purely machine learning approaches, our framework retains the volatility clustering and long-memory characteristics of FIGARCH, while incorporating predictive signals from FinBERT sentiment indicators and Random Forest-derived news impact probabilities. This design choice reflects the goal of bridging advanced NLP-based sentiment features with an established econometric framework, rather than replacing it with black-box methods. Nevertheless, recent state-of-the-art time-series models such as LSTMs, Transformers, and hybrid econometric–ML structures represent complementary directions. Future work could benchmark our framework against such alternatives or explore richer embedding-based representations, thereby extending the scope of our findings without diminishing the distinct contribution of this paper.

Acknowledgment

This research work was supported by the discovery grant RGPIN-2020-04782 from the Natural Sciences and Engineering Research Council of Canada (NSERC).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used `Grammarly` to check spelling, grammar, and word usage. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Appendix A. Supplementary Tables

Table A.4: Average model goodness-of-fit metrics (AIC, BIC, and LLF) for Stock and Cryptocurrency markets in the Open-to-Open time frame across different exogenous variable sets.

Market	**Exogenous Set**	**Average AIC**	**Average BIC**	**Average LLF**
Cryptocurrency	$\mathbf{X}^1$	-914.36	-884.08	466.18
	$\mathbf{X}^2$	-910.02	-879.73	464.01
	$\mathbf{X}^3$	-906.93	-876.65	462.47
	$\mathbf{X}^4$	-908.16	-877.88	463.08
	$\mathbf{X}^5$	-911.65	-861.17	470.82
	$\mathbf{X}^6$	-934.03	-897.02	478.02
	$\mathbf{X}^7$	-934.46	-883.99	482.23
	$\mathbf{X}^8$	-937.46	-873.53	487.73
	Base	-917.23	-897.01	464.62
Stock	$\mathbf{X}^1$	348.17	386.74	-165.09
	$\mathbf{X}^2$	28189.22	28227.78	-14085.61
	$\mathbf{X}^3$	4122.47	4161.03	-2052.23
	$\mathbf{X}^4$	4602.99	4641.55	-2292.49
	$\mathbf{X}^5$	546.07	610.34	-258.04
	$\mathbf{X}^6$	2641.35	2688.48	-1309.67
	$\mathbf{X}^7$	3624.03	3688.30	-1797.02
	$\mathbf{X}^8$	6752.55	6833.96	-3357.28
	Base	22104.68	22130.41	-11046.34

Table A.5: Average model goodness-of-fit metrics (AIC, BIC, and LLF) for Stock, Cryptocurrency, and Commodity markets in the Open-to-Close time frame across different exogenous variable sets.

Market	**Exogenous Set**	**Average AIC**	**Average BIC**	**Average LLF**
Commodity	$\mathbf{X}^1$	-1815.73	-1775.94	916.86
	$\mathbf{X}^6$	-2329.56	-2280.92	1175.78
	Base	2185.93	2212.47	-1086.96
Cryptocurrency	$\mathbf{X}^1$	-898.40	-868.11	458.20
	$\mathbf{X}^2$	-1200.46	-1170.17	609.23
	$\mathbf{X}^3$	-898.21	-867.93	458.11
	$\mathbf{X}^4$	-893.77	-863.49	455.89
	$\mathbf{X}^5$	-74.12	-23.65	52.06
	$\mathbf{X}^6$	-914.36	-877.34	468.18
	$\mathbf{X}^7$	-618.13	-567.66	324.07
	$\mathbf{X}^8$	-1200.99	-1137.06	619.50
	Base	-907.01	-886.79	459.50
Stock	$\mathbf{X}^1$	5673.89	5712.45	-2827.94
	$\mathbf{X}^2$	4936.47	4975.04	-2459.24
	$\mathbf{X}^3$	3807.30	3845.86	-1894.65
	$\mathbf{X}^4$	77210.99	77249.55	-38596.50
	$\mathbf{X}^5$	2288.21	2352.48	-1129.11
	$\mathbf{X}^6$	3420.73	3467.86	-1699.37
	$\mathbf{X}^7$	4201.82	4266.09	-2085.91
	$\mathbf{X}^8$	4473.58	4554.99	-2217.79
	Base	-2265.20	-2239.47	1138.60

Table A.6: Average model goodness-of-fit metrics (AIC, BIC, and LLF) for Stock and Cryptocurrency markets in the Close-to-Open time frame across different exogenous variable sets.

Market	**Exogenous Set**	**Average AIC**	**Average BIC**	**Average LLF**
Stock	$\mathbf{X}^1$	6039.89	6078.45	-3010.95
	$\mathbf{X}^2$	4116.86	4155.42	-2049.43
	$\mathbf{X}^3$	4326.77	4365.33	-2154.39
	$\mathbf{X}^4$	1640.01	1678.58	-811.01
	$\mathbf{X}^5$	5761.00	5825.27	-2865.50
	$\mathbf{X}^6$	8864.42	8911.55	-4421.21
	$\mathbf{X}^7$	17671.04	17735.31	-8820.52
	$\mathbf{X}^8$	2513.35	2594.76	-1237.68
	Base	6694.02	6719.75	-3341.01
Cryptocurrency	$\mathbf{X}^1$	1632.18	1662.47	-807.09
	$\mathbf{X}^2$	-3743.14	-3712.85	1880.57
	$\mathbf{X}^3$	-3741.46	-3711.18	1879.73
	$\mathbf{X}^4$	-3742.95	-3712.67	1880.48
	$\mathbf{X}^5$	-3730.44	-3679.97	1880.22
	$\mathbf{X}^6$	-3739.26	-3702.24	1880.63
	$\mathbf{X}^7$	-3732.58	-3682.10	1881.29
	$\mathbf{X}^8$	-3723.48	-3659.55	1880.74
	Base	-3766.38	-3746.16	1889.19

Table A.7: Average model goodness-of-fit metrics (AIC, BIC, and LLF) for Stock and Cryptocurrency markets in the Close-to-Close time frame across different exogenous variable sets.

Market	**Exogenous Set**	**Average AIC**	**Average BIC**	**Average LLF**
Stock	$\mathbf{X}^1$	26684.01	26722.57	-13333.01
	$\mathbf{X}^2$	-180.56	-142.00	99.28
	$\mathbf{X}^3$	14906.42	14944.98	-7444.21
	$\mathbf{X}^4$	4859.20	4897.76	-2420.60
	$\mathbf{X}^5$	621.37	685.64	-295.68
	$\mathbf{X}^6$	7878.49	7925.62	-3928.24
	$\mathbf{X}^7$	2623.56	2687.83	-1296.78
	$\mathbf{X}^8$	333.93	415.34	-147.97
	Base	143211.75	143237.47	-71599.88
Cryptocurrency	$\mathbf{X}^1$	-898.41	-868.12	458.20
	$\mathbf{X}^2$	4924.45	4954.74	-2453.23
	$\mathbf{X}^3$	-898.31	-868.02	458.15
	$\mathbf{X}^4$	-893.79	-863.50	455.89
	$\mathbf{X}^5$	1127.77	1178.24	-548.88
	$\mathbf{X}^6$	-914.39	-877.38	468.20
	$\mathbf{X}^7$	-1137.08	-1086.61	583.54
	$\mathbf{X}^8$	-902.68	-838.75	470.34
	Base	-907.02	-886.80	459.51